

# Time Series Imputation — Metric Consensus

## What the Sources Agree On, and Where They Do Not

The survey recorded which metrics the imputation literature reports. This document asks a different question of the same sources: for each metric, do they mean the same thing by it? Two papers can both report an nRMSE and compute numerically different quantities, and two papers can both report an MAE and agree completely. The distinction decides whether published values can be compared at all.

Each entry below records the evidence base, the properties every source agrees on, and the points on which they diverge. Sources are the 54 algorithm papers of the metrics-per-algorithm reference, the 11 metric-focused papers of the paper list (twelve entries, with VUS left out because it evaluates anomaly detection), and the five surveyed libraries. A paper counts towards a metric when it reports that metric on its own reconstruction, whatever heading the paper files it under, so SoftImpute’s held-out RMSE and ROSL’s recovery MAE count while a training loss alone does not. Where a claim rests on a single paper, that paper is named.

One metric carries two names, and the split is deliberate. The literature reports MAPE, so this document counts and analyses MAPE; the implementation uses sMAPE, because MAPE divides by the true value and explodes near zero on standardised data. Analysis therefore says MAPE throughout, and reported measurements say sMAPE.

### Consensus levels

- **Strong.** Every source defines and reads the metric the same way; published values are comparable.
- **Moderate.** The definition is agreed, but the sources differ on use, or share a property none of them discusses.
- **Contested.** Sources compute numerically different quantities under one name; values are not comparable without each paper’s definition.
- **Narrow.** Too few sources to establish consensus, whatever the metric’s standing elsewhere.

Table 1 collects the verdicts. *Algo* counts the 54 algorithm papers, *Metric* the 11 metric-focused papers and *Lib* the five libraries.

Table 1: Consensus verdict per metric, ordered as in the candidate set.

| Metric                                              | Algo | Metric | Lib | Verdict                                                                                                |
|-----------------------------------------------------|------|--------|-----|--------------------------------------------------------------------------------------------------------|
| <b>Pointwise Distance</b>                           |      |        |     |                                                                                                        |
| MAE                                                 | 28   | 7      | 4   | <b>Strong.</b> The one metric in the corpus with no conflicting attribution.                           |
| RMSE                                                | 20   | 9      | 4   | <b>Strong</b> on definition, with a limitation two metric papers formalise.                            |
| MSE                                                 | 20   | 3      | 2   | <b>Moderate.</b> Twelve papers train on it, six of them also report it, and none mentions the overlap. |
| MRE                                                 | 10   | 1      | 1   | <b>Contested.</b> Two papers compute a ratio of totals under the name of a mean.                       |
| MAPE                                                | 1    | 0      | 2   | <b>Narrow</b> for imputation. Surveyed as MAPE, implemented as sMAPE.                                  |
| nRMSE                                               | 4    | 0      | 0   | <b>Contested.</b> Four papers, four denominators, one of them never published.                         |
| ND                                                  | 1    | 0      | 0   | <b>Narrow.</b> One paper, whose definition is not in the published PDF.                                |
| <b>Distributional Divergence</b>                    |      |        |     |                                                                                                        |
| WD                                                  | 1    | 3      | 0   | <b>Strong</b> among its users, on a base of four papers.                                               |
| JSD                                                 | 1    | 2      | 0   | <b>Strong</b> among its users, always reported beside WD or KLD.                                       |
| KLD                                                 | 1    | 1      | 0   | <b>Moderate.</b> Common as a training term, reported as a metric once.                                 |
| <b>Temporal Structure</b>                           |      |        |     |                                                                                                        |
| ACF                                                 | 0    | 2      | 0   | <b>Narrow.</b> No algorithm paper reports it; both sources are metric papers.                          |
| DTW                                                 | 1    | 1      | 0   | <b>Narrow,</b> with a definition nobody disputes and two opposite motivations.                         |
| sMAE                                                | 1    | 0      | 0   | <b>Narrow.</b> One paper, which introduces it.                                                         |
| <b>Statistical Agreement</b>                        |      |        |     |                                                                                                        |
| Pearson                                             | 0    | 0      | 1   | <b>Narrow.</b> The only metric here that reaches practice through a library alone.                     |
| MI                                                  | 0    | 1      | 1   | <b>Narrow.</b> Same route as Pearson, same single library.                                             |
| $R^2$                                               | 1    | 1      | 0   | <b>Narrow.</b> Its imputation reading differs from its regression reading.                             |
| TOST                                                | 0    | 1      | 0   | <b>Narrow,</b> and not a ranking: it returns a verdict against a margin.                               |
| B-A                                                 | 0    | 1      | 0   | <b>Narrow,</b> and not a scalar: its one source uses it as a plot.                                     |
| CDT                                                 | 0    | 1      | 0   | <b>Narrow,</b> and its category is disputed between its source and this work.                          |
| <b>Set aside: probabilistic and domain-specific</b> |      |        |     |                                                                                                        |
| CRPS                                                | 6    | 0      | 1   | <b>Strong</b> among probabilistic papers. Equals MAE on a point estimate.                              |
| NLL                                                 | 3    | 0      | 0   | <b>Strong</b> among probabilistic papers. Monotone in RMSE on a point estimate.                        |
| PFC                                                 | 1    | 0      | 0   | <b>Narrow.</b> Defined for categorical data, with no continuous analogue.                              |

# 1 Pointwise Distance

These seven measure the distance between an imputed value and the true value at the same position, one position at a time. They differ in how they weight an individual error and in what, if anything, they divide by.

## MAE — Mean Absolute Error

**Consensus:** Strong. 28 algorithm papers, 7 metric papers, 4 libraries.

### Agreed across all sources

- Linear penalty: an error of twice the size contributes twice as much, so no single position can dominate the average.
- Reported in the unit of the data, which makes a value readable without a further transformation.
- More robust to outliers than the squared-error metrics, which is the property every source names first when it compares the two.
- The default basis for cross-study comparison, since it is the most frequently reported metric in the corpus.

### Where the sources differ

- *Loss and metric at once.* Seven papers train on MAE and report it as an imputation metric: SAITS, GRIN, SSGAN, BRITS, PoGeVon, CSAI and ReCTSi. BitGraph trains on a masked MAE too, but imputes nothing and reports MAE for forecasting. None of the eight mentions that the reported number is the objective the model minimised. MPIN is a ninth case and a different one: it trains on MSE and selects on MAE, so the metric it reports is the one it chose the model state with; the one it minimised goes unreported.
- *Sole metric by choice.* DeepMVI reports MAE alone across ten structurally heterogeneous datasets and gives no reason for it; RMSE is mentioned once in that paper, in passing.
- *Model selection separated from reporting.* MPIN selects on MAE over a held-out split and reports MAE and MRE, the only paper in the corpus that separates the two criteria explicitly.
- *Relegated for space.* SSGAN reports RMSE in the body and MAE in the appendix, stating that MAE shows similar trends and space is limited. The choice is editorial.

## RMSE — Root Mean Squared Error

**Consensus:** Strong on definition. 20 algorithm papers, 9 metric papers, 4 libraries. Two metric papers formalise a limitation.

### Agreed across all sources

- Quadratic penalty: a deviation twice as large contributes four times as much before the average is taken.
- Reported in the unit of the data, unlike MSE, because of the square root.
- More sensitive to large errors than MAE. Sources treat this as a diagnostic strength or as a weakness depending on what they want from it, but they agree on the mechanism.
- Zero algorithm papers train on RMSE directly, which makes it the most-reported metric that is never an objective.

### Where the sources differ

- *Headline by default.* SSGAN reports RMSE in the body and relegates MAE to the appendix, on the stated grounds that MAE shows similar trends and space is limited. The paper gives no reason to prefer the quadratic penalty, so the most-reported metric in that paper is the one that fitted the page.
- *Justified only as common.* M-RNN gives “common” as its whole justification, and scales all measurements to  $[0, 1]$ , so its reported values are not in clinical units despite the clinical setting.
- *Two errors, reported apart.* SoftImpute separates fitting error on observed entries from prediction error on held-out entries, which probes regularisation directly; most papers report one number.
- *Counted twice, once.* DynaMMo appears in this count and in MSE’s, because its one imputation figure is defined as a mean squared error in the text and captioned “Average rmse”. The paper reports a single quantity under two names, so no arithmetic separates them.

- *Formalised limitations.* Boursalie et al. show that a model producing a clearly wrong output distribution still reports a competitive RMSE. Dolatpour Fathkouhi et al. name the complementary case, where a low RMSE coexists with a reconstruction that misses the shape of a transient event. These are the only two places in the corpus where a metric’s limitation is demonstrated; everywhere else it is assumed.

## MSE — Mean Squared Error

**Consensus:** Moderate. 20 algorithm papers, 3 metric papers, 2 libraries.

### Agreed across all sources

- Quadratic penalty without the square root, so values are in squared units and are read comparatively.
- The dominant training objective of the deep learning literature. Twelve algorithm papers train on it: MOMENT, NuwaTS, Timer, UniTS, MPIN, TimesNet, NAOMI, M-RNN, SSSD, MTSIT, FGTI and LSCD.
- Amplifies a single bad position more than RMSE does, since the square is not undone.

### Where the sources differ

- *The undiscussed overlap.* Six papers both train on MSE and report it as an imputation metric: MOMENT, NuwaTS, Timer, UniTS, TimesNet and SSSD. Not one of them addresses what it means to report the quantity the model was optimised to minimise. This is the clearest gap in the corpus, and it is a gap of silence: no paper disagrees, because no paper raises it.
- *Naming drift.* DynaMMo defines its error as a mean squared error in the text and captions the corresponding figure “Average rmse”, on the same page. NAOMI reports what it calls an L2 loss where other papers write MSE.
- *Metric inheritance.* SAITS adds MSE for one comparison purely because the baseline it compares against reported it. NuwaTS reports MSE and MAE throughout and switches to RMSE and MAE for one appendix table, because that experiment follows the SAITS pipeline.
- *Natural for a Gaussian model.* DynaMMo justifies MSE as the counterpart of its EM procedure, which is the only justification in the corpus that ties the metric to the model’s own assumptions.

## MRE — Mean Relative Error

**Consensus:** Contested. 10 algorithm papers, 1 metric paper, 1 library.

### Agreed across all sources

- Normalises each error by the magnitude of the true value, so the result is dimensionless and comparable across sensors of different physical scales.
- Introduced when a dataset mixes variables of different magnitudes. Every reporting paper — MPIN, SAITS, GRIN, BRITS, SSSD, PoGeVon, FIM- $\ell$ , ReCTSi, SSD-TS and ST-MVL — works on multi-scale sensor data.
- Preferred over MAPE in the imputation literature, which reports MRE ten times against MAPE once. Richter et al. is the only metric paper that names it, and lists it beside MAE and RMSE without a definition.

### Where the sources differ

- *The name does not match the formula.* BRITS and ST-MVL both compute the sum of absolute errors divided by the sum of true values, which is a ratio of totals. A ratio of totals weights large-magnitude positions more heavily; a mean of ratios weights every position equally. The two are different quantities reported under one name.
- *Undefined at zero.* No source addresses the case where a true value is zero, which occurs routinely on standardised data. The limitation is well known outside this corpus and unmentioned inside it.
- *Defined by attribution.* GRIN names its metrics without defining them and attributes MRE to BRITS, so the formula must be recovered from the cited paper.

## MAPE — Mean Absolute Percentage Error

**Consensus:** Narrow for imputation. 1 algorithm paper, 0 metric papers, 2 libraries.

### Agreed across all sources

- Expresses each error as a percentage of the true value and averages those percentages, which reads directly as “imputed values are on average  $X$  % away from the truth”.
- Scale-invariant in the same sense as MRE, and motivated the same way when a dataset mixes magnitudes.

#### Where the sources differ

- *Chiefly a forecasting metric here.* TIDER is the only algorithm paper reporting MAPE for imputation. GPT4TS and BitGraph report it for forecasting, which is a different task with a different error profile. No metric paper names MAPE at all, so it reaches the corpus through two libraries and one algorithm.
- *Undefined at zero, again unaddressed.* The same gap as MRE, and with the same consequence on standardised data.

#### From MAPE to sMAPE at the implementation stage

- The metric the papers report is MAPE, and every count and every attribution above refers to MAPE. We implement sMAPE in its place, whose denominator is the mean of  $|y|$  and  $|\hat{y}|$  rather than  $|y|$  alone, because MAPE divides by the true value and therefore explodes near zero on the standardised series used throughout.
- The consequence is that values reported in the experiments are sMAPE values and cannot be set beside published MAPE results. Wherever this thesis surveys or ranks the literature it records MAPE; wherever it reports a measurement it reports sMAPE.

### nRMSE — Normalised Root Mean Squared Error

**Consensus:** Contested. 4 algorithm papers, 0 metric papers, 0 libraries.

#### Agreed across all sources

- The idea: divide RMSE by a reference magnitude so that the result is dimensionless and comparable across variables or datasets.
- The reading: zero is perfect and larger values are proportionally worse.

#### Where the sources differ

- *Every paper picks a different denominator.* IterativeSVD divides by the average data value in the complete dataset, and says why: to compare estimation accuracy between datasets. MissForest divides by the variance of the true values inside the square root, which makes its value an error relative to the data’s own standard deviation. CATSI divides by the range of each analyte within each patient and calls the result a normalised RMSD rather than an nRMSE. TRMF reports an nRMSE but defers the definition to an appendix that is not part of the published PDF.
- *Consequence.* Four papers report a metric of this name and no two of the values can be compared. Three of the four denominators are recoverable from the papers; the fourth is not recoverable at all.
- *Only CATSI states a purpose.* Its 13 clinical analytes span very different scales, so normalisation makes error rates comparable across them. No justification is given for the particular denominator.

### ND — Normalised Deviation

**Consensus:** Narrow. 1 algorithm paper, 0 metric papers, 0 libraries.

#### Agreed across all sources

- Summed absolute error divided by summed absolute true values, which makes it the  $L_1$  counterpart of nRMSE.

#### Where the sources differ

- *One source, no published definition.* TRMF reports ND beside nRMSE for both imputation and forecasting, and defers both definitions to the same unpublished appendix. The formula above is the standard one, not one this corpus confirms.
- *Also a forecasting metric.* GPT4TS reports ND for zero-shot forecasting, so the metric circulates beyond its one imputation source.

## 2 Distributional Divergence

These three compare the distribution of imputed values against the distribution of the truth, ignoring the order in which values occur. They catch the case where individual values are approximately right and the shape of the distribution is wrong.

### WD — Wasserstein Distance, or Earth Mover’s Distance

**Consensus:** Strong among its users. 1 algorithm paper, 3 metric papers, 0 libraries.

#### Agreed across all sources

- Measures the minimum cost of transporting one distribution onto the other, so the distance between values in the value space counts: moving mass from 1 to 2 is cheaper than moving it from 1 to 10. KLD and JSD have no such geometry.
- Stays finite when the two distributions do not overlap, which is where KLD fails.
- Can be computed without a ground truth, by comparing the imputed distribution against the observed one. This is what makes it the only metric in the set usable in a real deployment.

#### Where the sources differ

- *Proposed for the no-ground-truth case.* Farjallah et al. are the only source to make that argument, and validate it on Brussels and Madrid traffic data, where WD rankings agree with RMSE and MAE rankings.
- *Used as a training signal.* Huang et al. minimise the Wasserstein distance through optimal transport, so their reported WD confirms the objective the model was trained on.
- *Reported under a different name.* SST-GCN reports it as an earth mover’s distance, which is the same quantity. It normalises the value because the distance is unbounded.
- *Reverses a ranking.* Díaz-González et al. score BRITS against a random forest with MAE, RMSE, WD and TOST. The random forest wins on the pointwise pair and loses on the distributional pair, and the paper reports both without saying which should decide.

### JSD — Jensen–Shannon Divergence

**Consensus:** Strong among its users. 1 algorithm paper, 2 metric papers, 0 libraries.

#### Agreed across all sources

- Symmetric and bounded, which is how every source that uses it introduces it, and the reason each gives for preferring it to KLD.
- Numerically stable where the two distributions do not overlap, since it compares each against their mixture rather than against each other.
- Never reported alone. Both sources pair it with WD or with KLD.

#### Where the sources differ

- *No source argues JSD against WD.* Farjallah et al. propose both for the no-ground-truth case and give no reason to prefer either. Boursalie et al. include it as one step of a six-step framework, also without comparison.
- *Two different quantities under one abbreviation.* Boursalie et al. report a Jensen–Shannon *distance*, written as the mean of two Kullback–Leibler terms against the mixture, where Farjallah et al. report the divergence. The distance is the square root of the divergence, so the two are monotonically related and rank identically, but their values are not the same number.
- *Farjallah et al. specify no discretisation.* JSD on continuous data requires one and the value depends on it, so the reported figures cannot be reproduced from that paper. Boursalie et al. do describe how their distributions are formed.

### KLD — Kullback–Leibler Divergence

**Consensus:** Moderate. 1 algorithm paper as a metric, 1 metric paper, 0 libraries.

#### Agreed across all sources

- Measures the information lost in approximating one distribution with another, and is asymmetric, so the order of the arguments matters.
- Sensitive to the tails: a region where the reconstruction assigns little mass and the truth assigns much produces a large value.
- Infinite where the reconstruction assigns zero probability and the truth does not, which is the stated reason for preferring JSD.

#### Where the sources differ

- *Usually a training term.* GP-VAE, PoGeVon, TimeCIB and DMM all carry a KL term in an ELBO, where it regularises a latent posterior and says nothing about the imputation output. Counting those papers as KLD users would misread the corpus.
- *One paper reports it as a metric.* SST-GCN trains on the KL divergence between estimated and true speed distributions and evaluates with the same quantity, which makes the reported number a reading of its own objective.
- *A stated reason for the whole category.* SST-GCN is the only paper in the corpus that explains why it reports distributional metrics at all: it imputes a distribution rather than a point, so “standard regression metrics such as MAE and MAPE are not applicable”.

## 3 Temporal Structure

These three ask whether the reconstruction preserves the shape of the series over time. A reconstruction can have the right values and the right distribution and still be too smooth, misaligned, or stripped of its periodicity.

### ACF — Autocorrelation Agreement

**Consensus:** Narrow. 0 algorithm papers, 2 metric papers, 0 libraries.

#### Agreed across all sources

- Compares the correlation of the series with lagged copies of itself, which is a property no pointwise or distributional metric sees.
- Is a profile over lags, so using it as a metric requires a summary — typically a distance between two ACF profiles — and the choice of summary is part of the metric.

#### Where the sources differ

- *Both sources are metric papers.* Autocorrelation enters this work through the proposal bibliography and is measured zero times in the 54 algorithm papers, which makes it the clearest case of a metric the evaluation literature recommends and the algorithm literature ignores.
- *A diagnostic in one source, a score in the other.* Huang et al. read the autocovariance functions estimated from the imputed series off a plot and compare them by eye against the pre-specified values their method constrains. Díaz-González et al. instead compute an error between the two autocorrelation functions at lag 168 and report its distribution across variables together with per-station heatmaps, which makes theirs the only ACF value in the corpus that could be used to rank anything.
- *It agreed with the pointwise pair where it was tested.* Díaz-González et al. find that the random forest preserves autocorrelation better than BRITS on most variables, which is the same direction MAE and RMSE point in that paper. The metrics that reverse the ranking there are the Wasserstein distance and TOST, so ACF is not by itself evidence that a temporal metric sees what the pointwise ones miss.

### DTW — Dynamic Time Warping

**Consensus:** Narrow. 1 algorithm paper, 1 metric paper, 0 libraries.

#### Agreed across all sources

- Warps the time axis before measuring distance, so a pattern reproduced with a small shift still scores well. No other metric in the set tolerates a shift.

- Sensitive to shape and forgiving of timing, which makes it the natural counterpart to RMSE: a reconstruction that smooths through a transient scores well on RMSE and badly on DTW.
- Quadratic in series length, which is the reason every source gives, explicitly or implicitly, for its limited adoption.

#### Where the sources differ

- *Reported by one algorithm paper, and for a specific reason.* HKMF-T recovers blackouts, where a contiguous block must have the right shape and not merely the right average, so it reports an average DTW distance beside an average RMSE. It justifies the choice only as a frequently used elastic distance measure, and it is the only paper in the corpus reporting a temporal metric its own objective does not contain.
- *Argued for and used as the morphological metric.* Dolatpour Fathkouhi et al. name DTW as the answer to the failure they formalise, and report it in every results table beside RMSE, bias, empirical standard error and mean absolute relative difference. Their glucose results are the clearest demonstration in the corpus that the two disagree: two models within a few points of each other on RMSE differ by more than 90 on DTW, and one method holds its DTW stable across scenarios where the other degrades from 228.89 to 249.98.
- *The two sources use it for opposite reasons.* HKMF-T reports DTW because a shift should be forgiven, since a recovered blackout may sit slightly off in time and still be right. Dolatpour Fathkouhi et al. report it because a shape should be punished, since a smoothed reconstruction has no transient to align. The metric serves both readings, and neither source acknowledges the other.

### sMAE — Spectral Mean Absolute Error

**Consensus:** Narrow. 1 algorithm paper, 0 metric papers, 0 libraries.

#### Agreed across all sources

- Compares normalised power spectral densities, so it measures frequency content and is unaffected by amplitude.
- Needs the whole series, which it shares with ACF and DTW.

#### Where the sources differ

- *One paper, which introduces it.* LSCD computes it between Lomb–Scargle periodograms with missing values masked, and is the only source in the corpus to argue for combining a time-domain and a frequency-domain measure so that value accuracy and spectral preservation are both reported.
- *Only meaningful with periodic structure.* A series without periodicity gives a spectrum with nothing to compare, so the metric is agnostic in form and conditional in practice.
- *An unverifiable claim in the same paper.* LSCD’s abstract claims a CRPS improvement and no CRPS value appears anywhere in the paper.

## 4 Statistical Agreement

These six ask how the reconstructed values relate to the true ones as a pair of variables. How far apart they are is a separate question, and the pointwise metrics already answer it. All six come from the metric-focused literature or from a library. None is reported by any algorithm paper except  $R^2$ , once.

### Pearson — Pearson Correlation

**Consensus:** Narrow. 0 algorithm papers, 0 metric papers, 1 library.

#### Agreed across all sources

- Measures linear association, so it reads whether the reconstruction moves with the truth; where it sits is invisible to it.
- Invariant to any positive-slope affine transform, which is the property that makes it complementary to a distance and also its blind spot: a reconstruction shifted by a constant correlates perfectly with the truth.

#### Where the sources differ

- *It reaches practice through tooling alone.* ImputeGAP implements it, and its paper names two metrics in its own example, RMSE and MI, without listing Pearson among them. No algorithm paper and no metric paper reports it, so the implementation is the only source there is.
- *Undefined on a constant series.* That one source does not address the case where the reconstruction has zero variance, which is what a mean-fill produces on a blackout.

## MI — Mutual Information

**Consensus:** Narrow. 0 algorithm papers, 1 metric paper, 1 library.

### Agreed across all sources

- Measures shared information without assuming a functional form, so it detects nonlinear association that Pearson misses.

### Where the sources differ

- *Almost the same route to practice as Pearson.* It is implemented by ImputeGAP and reported by no algorithm paper. Apart from CRPS, which PyPOTS ships, the two are the only non-pointwise metrics any of the five libraries carries. They differ in one respect: the ImputeGAP paper names mutual information in its own worked example, beside RMSE, and names Pearson nowhere, so MI has a paper behind it and Pearson has only the code.
- *Binning is unspecified and decides the value.* On continuous data mutual information requires a discretisation, and no source states one, so no published value can be reproduced.
- *Invariant to any invertible map.* This is stronger than Pearson’s invariance and is not discussed by either source, though it means a bias and a rescaling both leave the metric almost unchanged.

## $R^2$ — Coefficient of Determination

**Consensus:** Narrow. 1 algorithm paper, 1 metric paper, 0 libraries.

### Agreed across all sources

- Reports the fraction of variance in the truth captured by the reconstruction, against a baseline that predicts the mean.
- Reads against fixed reference points: 1 is perfect, 0 matches the mean predictor and a negative value is worse than predicting the mean.
- Dimensionless, so it needs no normalisation to compare datasets.

### Where the sources differ

- *Reported for compatibility.* FIM- $\ell$  reports  $R^2$  in an appendix alongside MAE, RMSE and MRE, and does so for baseline compatibility.
- *The baseline means something different here.* In regression the comparison is against a fitted model; in imputation it is against a constant. A high  $R^2$  therefore says the method captured variation within the gap, which is a stronger claim than it sounds and one no source spells out.

## TOST — Two One-Sided Tests

**Consensus:** Narrow. 0 algorithm papers, 1 metric paper, 0 libraries.

### Agreed across all sources

- Tests whether the reconstruction and the truth are equivalent within a stated margin, rather than whether they differ.

### Where the sources differ

- *It is not a ranking.* TOST returns a  $p$ -value against a margin the user supplies, so two reconstructions can both be declared equivalent and the test will not order them. Using it as a metric means reading the binding  $p$ -value as a score, which is not what the procedure was built for.

- *The margin is a free parameter.* No source states one, and the verdict changes with it, so a TOST result is not interpretable without the margin that produced it.
- *Its one appearance is decisive.* Díaz-González et al. use it alongside WD to conclude that BRITS preserves the distribution better than the random forest that beats it on MAE and RMSE.

## Bland–Altman — Bias and Limits of Agreement

**Consensus:** Narrow. 0 algorithm papers, 1 metric paper, 0 libraries.

### Agreed across all sources

- Reports the mean difference between two measurements together with the interval containing most of the differences, so it separates systematic offset from spread.

### Where the sources differ

- *It is a plot.* Bland–Altman analysis is graphical by construction. Reducing it to the mean difference and the limit of agreement discards the pattern the plot exists to show, namely whether the disagreement varies with the magnitude of the measurement.
- *Two numbers to combine.* Any ranking use has to decide how to combine bias and spread, and no source does.
- *Its one source uses it exactly as a plot.* Poette et al. plot Bland–Altman and residual diagrams for mean blood pressure to read agreement across the physiological range, and report increased variance and bias at the clinically sensitive ends. That reading is the pattern across magnitudes, which is precisely what a scalar summary would throw away.

## CDT — Cohen’s Distance Test

**Consensus:** Narrow, with a disputed category. 0 algorithm papers, 1 metric paper, 0 libraries.

### Agreed across all sources

- A standardised difference in means: the gap between the two means divided by their pooled standard deviation, which makes it dimensionless.

### Where the sources differ

- *Its source files it as a statistical distance; this work does not.* Boursalie et al. place CDT in a row group labelled “statistical distance”, together with the Kullback–Leibler and Jensen–Shannon divergences, which reads it as a measure of distributional dissimilarity. Cohen’s  $d$  compares centres and says nothing about the shape of the distribution around them, so it is filed here under statistical agreement, alongside Bland–Altman, which measures the same thing without standardising it.
- *It duplicates Bland–Altman.* Both reduce to a difference in means, and both read zero for any reconstruction that preserves the mean, whatever else it does to the series.

## 5 Set Aside: Probabilistic and Domain-Specific

These three sit outside the working set of nineteen, for two different reasons. CRPS and NLL are computed by no experiment here, because on a point estimate each collapses onto a metric the set already holds. PFC is computed on every reconstruction and left out of the analysis instead, because the quantity we compute is our own adaptation of MissForest’s categorical metric. The reasons are worth recording, since two of the three are well established in the corpus.

## CRPS — Continuous Ranked Probability Score

**Consensus:** Strong among the probabilistic papers. 6 algorithm papers, 0 metric papers, 1 library.

### Agreed across all sources

- A proper scoring rule: it rewards a sharp distribution and a calibrated one at once, so widening the prediction to cover uncertainty does not improve the score.
- Scores the whole predictive distribution, and reaches zero only for a point mass at the true value.

- The common ground for comparing probabilistic methods. BayOTIDE, PriSTI, CSDI, CSBI, SSD-TS and FGTI all report it for that purpose, the last of them only in an appendix.

#### Where the sources differ

- *Used as a bridge.* BayOTIDE reports RMSE and MAE beside CRPS and NLLK so its probabilistic output can be set against deterministic baselines. SSD-TS goes further and splits its evaluation explicitly by whether a model can quantify uncertainty, adding CRPS only for those that can.
- *The collapse is never stated.* For a point estimate the predictive distribution is a point mass and CRPS reduces exactly to the absolute error, so  $\text{CRPS} = \text{MAE}$ . None of the six papers says so, though all six report both.
- *Claimed without being reported.* LSCD’s introduction claims a CRPS improvement over time-domain baselines. The word appears once in the paper and no CRPS value is reported anywhere.

## NLL — Negative Log-Likelihood

**Consensus:** Strong among the probabilistic papers. 3 algorithm papers, 0 metric papers, 0 libraries.

#### Agreed across all sources

- Reads the probability the model assigns to the values that actually occurred, so a lower value means the posterior explains the data better.
- Applies only to a model that outputs a distribution. GP-VAE, BayOTIDE and TimeCIB are the three that report it.
- Penalises confident errors far more sharply than CRPS does, since the log of a small probability grows without bound.

#### Where the sources differ

- *The restriction is stated once.* TimeCIB restricts NLL to VAE-based baselines, marks the restriction in its tables and gives three reasons for excluding the others. The same restriction is implicit everywhere else and is written down nowhere else.
- *An undefined variant.* BayOTIDE reports an NLLK without defining the  $K$  anywhere in the paper.
- *The collapse, again unstated.* Estimating a Gaussian scale from the residuals of a deterministic predictor gives  $\sigma = \text{RMSE}$ , which makes NLL a monotone function of RMSE and therefore a metric that produces the identical ordering. No source mentions it.

## PFC — Proportion of Falsely Classified Entries

**Consensus:** Narrow. 1 algorithm paper, 0 metric papers, 0 libraries.

#### Agreed across all sources

- Counts the fraction of categorical entries imputed with the wrong category, which has no continuous analogue.

#### Where the sources differ

- *One source, one data type.* MissForest defines PFC for the categorical half of a mixed-type dataset and reports NRMSE for the continuous half. The metric never appears again in the corpus.
- *The adaptation is ours.* This work implements a continuous version measuring the fraction of imputed values inside a 10 % relative tolerance band. It is serviceable, and because it departs from MissForest’s definition, its values cannot be set beside published PFC results.

## 6 What the Comparison Shows

Three patterns run across the twenty-two entries.

**Consensus is strong exactly where the metric is common, and nowhere else.** MAE and RMSE are the two most-reported metrics and the two whose definitions no source disputes. Below them, agreement decays with frequency: MSE is agreed but its dual role is unexamined, MRE and nRMSE are reported under one name for different quantities, and the remaining seventeen rest on one or two sources each, CRPS and NLL excepted.

Frequency and clarity coincide, which means the field has agreed on the metrics it uses most and has not had the argument about the rest.

**The disagreements that matter are about denominators.** Four papers report an nRMSE with four different reference quantities, one of which was never published; two papers report an MRE that is a ratio of totals rather than a mean of ratios. In both families the values look comparable, carry the same name, and are not comparable. This is a worse failure than an openly contested metric, because nothing on the surface signals it.

**The metrics with the clearest rationale are the least used.** The sources that argue hardest for a metric — Farjallah et al. for the Wasserstein distance, Dolatpour Fathkouhi et al. for shape-aware measurement, LSCD for spectral agreement — are answered by one algorithm paper each or by none. Five metrics appear in the metric-focused literature and in no algorithm paper at all, and Pearson appears in neither, reaching practice through a library alone. The critique of pointwise evaluation exists, is specific, and has not reached the papers that evaluate.